ELS63, LE910Q1 and LE310Q1 FOTA AT Script

Application Note

80798NT12167A Rev.1
2025-11-25
Released
Confidential

Technical Documentation

Contents

1 Applicability Table 3
2 Introduction 4
2.1 Scope 4
2.2 Audience 4
2.3 Contact Information, Support 4
2.4 Conventions 4
2.5 Terms and Conditions 5
2.6 Disclaimer 5
3 Module Groups Classification 6
4 How to Upgrade the Firmware of a Telit Cinterion Module 7
4.1 How to download the Delta-FW file from an FTP server 7
4.1.1 Procedure valid for Group E 7
4.2 How to restart the download of the Delta-FW file from an FTP server 8
4.2.1 Procedure valid for Group E 8
4.3 How to download the Delta-FW file from an HTTP server 8
4.3.1 Procedure valid for Group E 8
4.4 How to inject the Delta-FW file through a user's port connection 9
4.4.1 Procedure valid for Group E 9
4.5 How to deploy the Delta-FW 10
4.5.1 Procedure valid for Group E 10
5 AT Command Description 11
5.1 AT Commands for Group E 11
5.2 URC Sequence 11
5.2.1 Update result messages for Group E 11
5.3 Error Codes 11
5.3.1 Error Codes for Group E 11
6 Acronyms and Abbreviations 12
7 Related Documents 13
8 Document History 14

1 Applicability Table

Table 1: Applicability Table

Products
ELS63 Series
LE910Q1 Series
LE310Q1 Series

2 Introduction

2.1 Scope

Aim of this document is to describe how to upgrade the firmware of Telit Cinterion Modules using FOTA functionality via AT commands, either when a Telit Cinterion Module has to download the Delta-FW file from an FTP server (§4.1), from an HTTP server (§4.3) or when a Telit Cinterion Module can receive it through a user's serial connection (§4.4).

2.2 Audience

This document is intended for those users that need to upgrade via FOTA the firmware of their modules without direct control of FOTA servers.

2.3 Contact Information, Support

For technical support and general questions, e-mail:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use: https://www.telit.com/contact-us/

For Product information and technical documents, visit: https://www.telit.com

2.4 Conventions

Note:	Provide advice and suggestions that may be useful when integrating the module.
Danger:	This information MUST be followed, or catastrophic equipment failure or personal injury may occur.
Warning:	Alerts the user on important steps about the module integration.

All dates are in ISO 8601 format, that is YYYY-MM-DD.

2.5 Terms and Conditions

Refer to https://www.telit.com/hardware-terms-conditions/.

2.6 Disclaimer

THE MATERIAL IN THIS DOCUMENT IS FOR INFORMATIONAL PURPOSES ONLY. TELIT CINTERION RESERVES THE RIGHT TO MAKE CHANGES TO THE PRODUCTS DESCRIBED HEREIN. THE SPECIFICATIONS IN THIS DOCUMENT ARE SUBJECT TO CHANGE AT THE DISCRETION OF TELIT CINTERION WITHOUT PRIOR NOTICE. THIS DOCUMENT IS PROVIDED ON “AS IS” BASIS ONLY AND MAY CONTAIN DEFICIENCIES OR INADEQUACIES. TELIT CINTERION DOES NOT ASSUME ANY LIABILITY FOR INFORMATION PROVIDED IN THE DOCUMENT OR ARISING OUT OF THE APPLICATION OR USE OF ANY PRODUCT DESCRIBED HEREIN.

TELIT CINTERION GRANTS A NON-EXCLUSIVE RIGHT TO USE THE DOCUMENT. THE RECIPIENT SHALL NOT COPY, MODIFY, DISCLOSE, OR REPRODUCE THE DOCUMENT EXCEPT AS SPECIFICALLY AUTHORIZED BY TELIT CINTERION.

TELIT CINTERION AND THE TELIT CINTERION LOGO, ARE TRADEMARKS OF TELIT CINTERION AND ARE REGISTERED IN CERTAIN COUNTRIES. ALL OTHER REGISTERED TRADEMARKS OR TRADEMARKS MENTIONED IN THIS DOCUMENT ARE THE PROPERTY OF THEIR RESPECTIVE OWNERS AND ARE EXPRESSLY RESERVED BY TELIT CINTERION (AND ITS LICENSORS).

3 Module Groups Classification

In the following chapters we refer to the Telit Cinterion Modules of group E.

Group E	ELS63 Series, LE910Q1 Series, LE310Q1 Series ELS63-I, ELS63-IG, LE910Q1-WW, LE910Q1-WWG, LE910Q1-SN, LE910Q1-SNG, LE910Q1-SF, LE310Q1-WW, LE310Q1-SN, LE310Q1-EU

4 How to Upgrade the Firmware of a Telit Cinterion Module

This chapter reports the list of AT commands needed to upgrade the firmware of Telit Cinterion Modules. We report in black the AT commands sent to the module and in blue the expected AT answers.

The Delta-FW file can be both downloaded from an FTP Server or injected through a User's Port Connection.

4.1 How to download the Delta-FW file from an FTP server

In this chapter, we assume that the Telit Cinterion Module can download the Delta-FW file from an FTP server.

4.1.1 Procedure valid for Group E

```
AT#SCFG=1,1,1460,90,600,50
OK
// Change 3rd and 4th parameters with the ones required
// by your context activation.
// If your context activation doesn't require both parameters
// <userId> and <pwd>, please skip the input of both of them.
// Try max 3 times, otherwise retry after 15 minutes
AT#SGACT=1,1,"username","password"
#SGACT: x.y.w.z

OK

// The FTPCFG timeout can be changed for slow network
AT#FTPCFG=501,0,0,1
OK

// Change 1st, 2nd and 3rd parameter with the ones required by
// the FTP server you are going to connect to.
// Try max 3 times, otherwise retry after 5 minutes or
// restart with AT#SGACT
AT#FTPOPEN="<SERVER_ADDRESS>:<SERVER_PORT>","<USERNAME>",
"<PASSWORD>",1
OK

// Change the parameter with the right FTP path of the
// Delta-FW file in the FTP server you are connected to
```

```
AT#FTPCWD=" <FOLDER_NAME>"
OK
// Try max 3 times, otherwise retry after 5 minutes or
// restart with AT#FTPOPEN
AT#FTPGETOTA="ELS63-I_X.X.X-B010-B010A-P0Y.000000_delta.par",1
##############
OK

AT#FTPCLOSE
OK

AT#SGACT=1,0
OK
```

4.2 How to restart the download of the Delta-FW file from an FTP server

4.2.1 Procedure valid for Group E

It is not possible to restart a download process started by #FTPGETOTA, that was previously interrupted.

4.3 How to download the Delta-FW file from an HTTP server

In this chapter, we assume that the Telit Cinterion Module can download the Delta-FW file from an HTTP server.

4.3.1 Procedure valid for Group E

```
AT#SCFG=1,1,1460,90,600,50
OK

// Change 3rd and 4th parameters with the ones required
// by your context activation.
// If your context activation doesn't require both parameters
// <userId> and <pwd>, please skip the input of both of them.
// Try max 3 times, otherwise retry after 15 minutes
AT#SGACT=1,1,"username","password"
#SGACT: x.y.w.z

OK
```

```
AT#HTTPCFG=1,<SERVER_ADDRESS>,<SERVER_PORT>,1,"<USERNAME>",
"<PASSWORD>",0,120,1
OK
AT#HTTPQRY=1,0,
"/<FOLDER_NAME>/ELS63-I_X.X.X-B010-B010A-P0Y.000000_delta.par"
OK

#HTTPRING: 1,200,"",59536

AT#HTTPRCVOTA=1,1
##############
OK

AT#SGACT=1,0
OK
```

4.4 How to inject the Delta-FW file through a user's port connection

In this chapter, we assume that the Telit Cinterion Module can receive the needed Delta-FW file through a user's serial connection.

4.4.1 Procedure valid for Group E

```
// Send the command #OTAUPW to start the injection of the Delta-FW
// file into module's flash memory
AT#OTAUPW
CONNECT
// Send the Delta-FW file to the module via user's USB/SERIAL
// interface
// remind to use hardware flow control while sending the file
// Once the sending of the file has been terminated, close the
// connection through the escape sequence (+++).
+++
OK

// Alternatively, you can send the command #OTAUPW along with its
// 2 optional sub-parameters
AT#OTAUPW=0,<data size>
CONNECT
// Send the Delta-FW file to the module via user's USB/SERIAL
// interface
```

```
// remind to use hardware flow control while sending the file
// Once <data size> bytes have been uploaded to the module, the
// connection closes automatically and you don't need to send the
// escape sequence (+++).
OK
```

4.5 How to deploy the Delta-FW

Once the Delta-FW file has been downloaded/injected into Telit Cinterion Module, we send the following sequence of AT Commands to upgrade the FW of the Telit Cinterion Module itself.

4.5.1 Procedure valid for Group E

```
// Next AT Command is to ask the FOTA engine to start the
// validation/update process to upgrade the FW of a Telit Cinterion
Module
AT#OTAUP=0
OK

// URC to inform that FOTA engine has finished its work
#OTAEV: Module Upgraded To New Fw

// Send the next command to check that the firmware has been
// updated successfully
AT#SWPKGV
52.00.000-B010A-P0Y.000000
M0Y.000000-B001
P0Y.000000
A0Y.000000-B010A
OK
```

Note: The request for the FOTA engine to start the validation/update process to upgrade the FW of a Telit Cinterion Module is triggered also by module reset or power off/power on sequence.

5 AT Command Description

5.1 AT Commands for Group E

Refer to the related document [1]
"ELS63, LE910Q1and LE310Q1 AT Commands Reference Guide"
Document Code 80798ST11194A.

5.2 URC Sequence

5.2.1 Update result messages for Group E

Table 2: Update result messages

URC		Description
Success Case	#OTAEV: Module Upgraded to New Fw	Notification – the FOTA engine has finished its work successfully
Failure Case (currently not supported by ELS63 Series, LE910Q1 Series and LE310Q1 Series)	#OTAEV: OTA Fw Upgrade Failed	Notification – An error occurred while the FOTA engine was doing its work

5.3 Error Codes

5.3.1 Error Codes for Group E

No error codes.

6 Acronyms and Abbreviations

Table 3: Acronyms and Abbreviations

Acronym	Definition
ADC	Analog – Digital Converter
CA	Connection Agent
CLK	Clock
CMOS	Complementary Metal – Oxide Semiconductor
CS	Chip Select
DAC	Digital – Analog Converter
DTE	Data Terminal Equipment
ESR	Equivalent Series Resistance
EVB	Evaluation Board
EVK	Evaluation Kit
GPIO	General Purpose Input Output
HS	High Speed
HSDPA	High Speed Downlink Packet Access
HSIC	High Speed Inter Chip
HSUPA	High Speed Uplink Packet Access
I2C	Inter Integrated Circuit
I/O	Input Output
MISO	Master Input – Slave Output
MOSI	Master Output – Slave Input
MRDY	Master Ready
PCB	Printed Circuit Board
PDP	Packet Data Protocol
RSSI	Radio Signal Strength Information
RTC	Real Time Clock
SIM	Subscriber Identification Module
SPI	Serial Peripheral Interface
SRDY	Slave Ready
TATC	Telit AT Controller
TTSC	Telit Technical Support Centre
UART	Universal Asynchronous Receiver Transmitter
UMTS	Universal Mobile Telecommunication System
URI	Universal Resource Identifier
URC	Unsolicited Result Code
USB	Universal Serial Bus
VNA	Vector Network Analyzer
VSWR	Voltage Standing Wave Radio
WCDMA	Wideband Code Division Multiple Access

7 Related Documents

For current documentation and downloads, refer to https://dz.telit.com/.

Table 4: Related Documents

S.no	Doc Code	Document Title
1	80798ST11194A	ELS63, LE910Q1 and LE310Q1 AT Commands Reference Guide

8 Document History

Table 5: Document History

Revision	Date	Changes
1	2025-11-25	Changed title to ELS63, LE910Q1 and LE310Q1 FOTA AT Script Updated Applicability Table Updated Group E list in chapter 3 Added Note in chapter 4.5.1 Updated related document in chapter 5.1 and in Table 4 Updated Table 2 in chapter 5.2.1
0	2024-07-16	Initial release

From Mod.0817 Rev.10

© Telit Cinterion. All rights reserved.

Technical Documentation